Tanmay Bhuskute

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SUMMARY

Software Engineer and M.S. Computer Science candidate (Dec 2026) building full-stack apps with React, TypeScript, Node.js, and REST APIs backed by Zod contracts and relational databases, plus AWS-based backend automation. AI-native across the SDLC: planning, implementation, testing, and review.

EDUCATION

Master of Science in Computer Science Aug 2024 - Dec 2026
University of Cincinnati
Relevant Coursework: Advanced Algorithms, Deep Learning, Intelligent Data Analysis, Database Theory.

Bachelor of Engineering in Computer Engineering Aug 2018 - Jul 2022
AISSMS College of Engineering
Relevant Coursework: Data Mining, Data Analytics, Machine Learning, Object-Oriented Programming.

TECHNICAL SKILLS

Programming: Java, Python, JavaScript, TypeScript, C/C++, SQL, Go.

Web & Backend: React, Node.js, Express, Django, Flask, FastAPI, HTML/CSS, REST APIs, Zod.

Databases: PostgreSQL, MySQL.

Developer Tools: Git, GitLab CI/CD, Jenkins, AWS (S3, EC2, Lambda, IAM), Docker, Kubernetes, ArgoCD, Helm, Terraform, Linux, Vitest, SonarQube, Azure, GCP.

AI & Data: Gemini, Vertex AI, PyTorch, TensorFlow, scikit-learn, LangChain, pandas, NumPy.

WORK EXPERIENCE

Graduate Researcher —- University of Cincinnati Feb 2026 – Present

- Designed two-stage memory-augmented LLM pipeline decoupling video understanding from device automation for Android GUI bug reproduction, evaluated against **ViBR** (GPT-4o + GroundingDINO + CLIP baseline) across 53 runs on 20+ apps.
- Demonstrated **25.7x latency reduction** (39.4s vs 1012.1s avg per run) while preserving task-completion fidelity.

Software Development Engineer —- Siemens Industry Software Jul 2022 – Jun 2024

- Automated **Teamcenter X** infrastructure workflows with Python (Boto3), replacing legacy Ruby scripts and reducing deployment time by **40%** across AWS environments using IAM roles for secure access.
- Executed **44 production environment upgrades** in 4 months, diagnosing live infrastructure issues across Linux, EC2, databases, and Teamcenter X services.
- Designed **Ansible-based Blue-Green deployment** framework, reducing production upgrade downtime by **50%**.
- Migrated **40+ production environments** from process-based deployments to Kubernetes using ArgoCD, Helm, GitLab CI/CD, Amazon EFS, and HashiCorp Vault.
- Served as Quality Coach and SonarQube Specialist, increasing automated code coverage from **35% to 78%**; mentored **5 software engineers** through structured onboarding.

Software Developer Intern —- Siemens Industry Software Mar 2022 – Jun 2022

- Automated provisioning of AWS resources with **Terraform**, reducing deployment time by **30%**.
- Developed **Go + Terratest** framework, catching **95%** of misconfigurations pre-production.

Software Developer Intern —- Government of Maharashtra, Water Resource Department Sep 2021 – Dec 2021

- Built and deployed a full-stack **React and MySQL** workflow application that digitized departmental record management, automated manual processes, and improved operational efficiency by **40%**.

PROJECTS AND RESEARCH

PolicyPilot – Medical Benefit Policy Intelligence Portal [React, TS, Node, Express, Zod, Gemini] May 2026

- Built full-stack TypeScript app (React SPA + Express REST API) with **17 REST/MCP tool endpoints** validated by **Zod schemas** at every data boundary, eliminating a class of runtime type-mismatch bugs.
- Architected evidence-grounded RAG pipeline over Google Gemini 2.5-flash with citation-backed answers; added Vitest unit/integration tests and WCAG AA accessibility coverage.

EXTRACURRICULAR ACTIVITIES

- **Finalist, RevolutionUC 2025 Hackathon** (300 participants, 31 schools) | **SAFe Practitioner 5.1** (Mar 2023).